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 Studierichting: Informatica
 Oefeningenreeks 3G nr. 18.5

1. Domein: $\mathbb{R}$

2. Asymptoten:

Verticale asymptoten:

geen nulpunten in de noemer $\Rightarrow$ geen VA

Horizontale asymptoten:

$$\lim_{x \rightarrow +\infty} \sqrt[3]{x^3 - 3x + 2} = +\infty$$

$$\lim_{x \rightarrow -\infty} \sqrt[3]{x^3 - 3x + 2} = -\infty$$

geen horizontale asymptoten

Schuine asymptoot:

$$y = \sqrt[3]{x^3 - 3x + 2}$$

$$y^3 = x^3 - 3x + 2$$

$$y = ax + b \Rightarrow y^3 = (ax + b)^3$$

$$y^3 = a^3x^3 + 3a^2bx^2 + 3ab^2x + b^3$$

$$\begin{cases} a^3 = 1 \Rightarrow a = 1 \\ 3a^2b = 0 \Rightarrow b = 0 \end{cases}$$

$$y = x$$

Ligging ten opzichte van SA:

$$(f(x))^3 - y^3 = (x^3 - 3x + 2) - x^3 = -3x + 2$$

x	$\frac{2}{3}$		
$(f(x))^3 - y^3$	+	0	-
	boven	snijpunt	onder

3. Tekenonderzoek $f(x)$:

$$f(x) = 0 \Leftrightarrow x^3 - 3x + 2 = 0$$

$$(x - 1)^2(x + 2) = 0$$

$$x = 1 \text{ of } x = -2$$

4. Tekenonderzoek $f'(x)$:

$$f'(x) = \left((x^3 - 3x + 2)^{\frac{1}{3}} \right)'$$

$$f'(x) = \frac{1}{3}(x^3 - 3x + 2)^{-\frac{2}{3}}(3x^2 - 3)$$

$$f'(x) = \frac{x^2 - 1}{(x^3 - 3x + 2)^{\frac{2}{3}}}$$

$$f'(x) = \frac{(x - 1)(x + 1)}{((x - 1)^2(x + 2))^{\frac{2}{3}}}$$

$$f'(x) = \frac{x + 1}{(x - 1)^{\frac{1}{3}}(x + 2)^{\frac{2}{3}}}$$

$$f'(x) = 0 \Leftrightarrow x = -1$$

5. Tekenonderzoek $f''(x)$:

$$f''(x) = \left(\frac{x^2 - 1}{(x^3 - 3x + 2)^{\frac{2}{3}}} \right)'$$

$$f''(x) = \frac{2x(x^3 - 3x + 2)^{\frac{2}{3}} - (x^2 - 1) \left(\frac{2}{3}(x^3 - 3x + 2)^{-\frac{1}{3}}(3x^2 - 3) \right)}{(x^3 - 3x + 2)^{\frac{4}{3}}}$$

$$f''(x) = \frac{2x(x^3 - 3x + 2) - 2(x^2 - 1)^2}{(x^3 - 3x + 2)^{\frac{5}{3}}}$$

$$f''(x) = \frac{2x^4 - 6x^2 + 4x - (2x^4 - 4x^2 + 2)}{(x^3 - 3x + 2)^{\frac{5}{3}}}$$

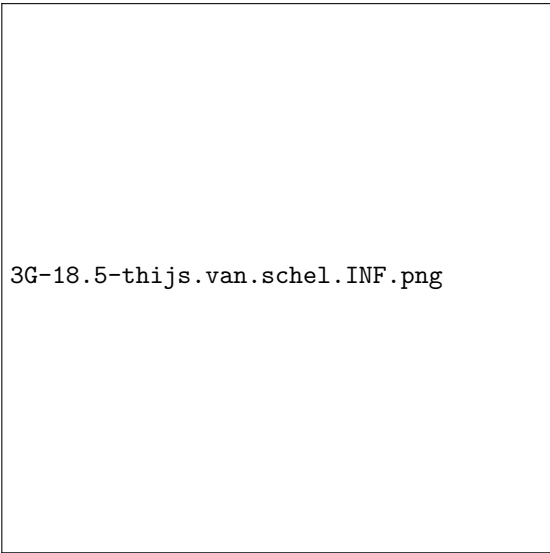
$$f''(x) = \frac{-2x^2 + 4x - 2}{((x - 1)^2(x + 2))^{\frac{5}{3}}}$$

$$f''(x) = \frac{-2(x - 1)^2}{(x - 1)^{\frac{10}{3}}(x + 2)^{\frac{5}{3}}}$$

$$f''(x) = \frac{-2}{(x-1)^{\frac{4}{3}}(x+2)^{\frac{5}{3}}}$$

6. Tekentabel:

x	-2			-1	1		
$f(x)$	$-$	0	$+$	$\sqrt[3]{4}$	$+$	0	$+$
$f'(x)$	$+$	$+\infty$	$+$	0	$-$	$\pm\infty$	$+$
$f''(x)$	$+$	$ $	$-$	$-$	$-$	$ $	$-$
$f(x)$	$\nearrow$	$BP(0)$	$\nearrow$	$M\left(\sqrt[3]{4}\right)$	$\searrow$	$m(0)$	$\nearrow$
	$\smile$		$\smile$		$\smile$		$\smile$



References